PAQ-KV: Supplementary Material

Anonymous submission

	LDU	MMLB
Documents	80	80
Complete queries at gate	463	444
Gold-bearing queries / doc	5.79	5.50
Median (mean) union pages	44 (45.0)	28 (36.5)

Table S1: The confirmatory evaluation pin. “Complete queries” are rows where every gated arm scored; the below-page gate completed all 907 pinned queries with zero skips, so no imputation was needed (the historical page-level run had 4 timeout-missing queries, immaterial under worst-case imputation).

This document provides the supplementary material for the submission “Prefill Once, Ask Many: Training-Free Reversible Below-Page KV Selection for Multi-Query Long-Document Visual Question Answering” (PAQ-KV). Plain-numbered references (§4.2, Table 1, Figure 3) point to the main paper; floats in this document are numbered S1, S2, …Notation follows the main paper: MMLB abbreviates MMLongBench-Doc, LDU abbreviates LongDocURL, paired contrasts are reported as $\Delta\ [l, u]$ with document-clustered bootstrap 95% confidence intervals, and LB_{95} is the interval’s lower bound.

Algorithm S1 PAQ-KV: per-query reversible below-page KV-block selection with a 4D restricted-attention mask

Require: document pages rendered as in the full-context arm ($\leq$64K tokens); queries $q_1, \ldots, q_k$; nominal budget $b = 5\%$; frozen VLM

prefill the document once; retain the full KV cache C (context length n_{ctx}); record page-aligned 64-token block spans $B_1, \ldots, B_m$ and the always-keep scaffold set $\mathcal{A}$
for each query q_j **do**
  forward the tokenized question over C; capture question-to-context attention via per-layer hooks; **crop** C back to n_{ctx}
  $\text{score}(B_i) \leftarrow$ aggregated question-row attention mass on block B_i’s positions
  $S_j \leftarrow$ top blocks by score at the nominal budget $b \cdot n_{\text{ctx}}$
  build the 4D additive mask M_j: decode rows may attend to $\mathcal{A} \cup S_j \cup q_j \cup$generated tokens; all other context positions get $-\infty$
  decode the answer from C under M_j; no gathering, so K/V positions and M-RoPE indices are unchanged
  discard M_j; C is intact for q_{j+1}
end for

A Implementation and Reproducibility

A.1 Pin Construction, Algorithm, and Gate Materials

Pin construction. For each document we build the union of the benchmark’s own-document pages across that document’s queries, dropping cross-document 64K padding, so every query’s evidence is present in one prefill. Documents qualify with at least three gold-bearing queries and at most 110 union pages, a bound that targets the nominal $\approx$64K-token context at read resolution (realized lengths reach $\approx$92K on the largest documents); selection is seed-0 frozen. Table S1 summarizes the confirmatory pin.

Hardware and software. All experiments ran on a single NVIDIA RTX PRO 6000 Blackwell GPU (96 GB) under Ubuntu 22.04, with Python 3.11.6, PyTorch 2.10.0 (CUDA 12.8), transformers 5.5.4, and peft 0.19.1. Models load in bfloat16, and the resolved revision hash of every model checkpoint is recorded in the released provenance manifests (§A.2).

Hyperparameters. Every setting was frozen before the gate that uses it; the only tuned selection in the study is the pre-registered, development-only InternVL tuning study of §B.4.

- Nominal evidence budget $b = 5\%$, the pre-registered gate value; swept at $\{2.5, 10\}\%$ on the full pin (Table S6) and at $\{5, 10, 20, 30\}\%$ in the InternVL tuning study.
- KV block size 64 tokens, page-aligned: fixed a priori, no sweep; the page-granularity comparison of §C.1 probes the coarser limit.
- Block scorer: mean attention over all layers, heads, and question rows (Equation 1 of the main paper). Alternatives evaluated and not adopted: max aggregation (Table S7), a single sharp expert head, and a learned per-head mixture (§C.4).

- Commit-once aggregate $M = \min(3, k)$ first queries: fixed a priori; sensitivity checks remain open (§D).
- Context target $\approx$64K tokens nominal (realized up to $\approx$92K) and page rendering at 144 DPI, both fixed by the pin protocol above.
- Decoding: greedy (do_sample=False), $\text{max\_new\_tokens} = 128$, identical across all arms.
- Attention implementation: SDPA for prefill and decode; eager attention only during the per-query preview, so question-to-context attention can be captured by hooks.
- Seeds: 0 for pin selection, the random-selection arms, and the bootstrap; the bootstrap uses 10,000 document-clustered percentile draws.

A.2 Reproducibility and Protocol Notes

Pre-registration discipline. Every gate in this study was frozen (bars, comparators, budgets, imputation policy) before its run: the two-comparator isolation gate; the PAQ-T accuracy-evaluation bars (non-inferiority/improvement/inferior margins, per-cell power floors, adversarial missingness imputation); the per-head-mixture transfer gate; and the PAQ-KV decode-identity kill criterion, screen gates, per-cell improvement bars, and sharp-scorer (V1) gate. Two early positives did not survive adversarial review and were retracted before this draft: a first isolation gate against the eviction family (invalidated by a weak-static artifact and replaced by the two-comparator gate), and an early "outperforms the gold-page oracle" claim (corrected to "matches" at power). The commitment-pressure mechanism hypothesis failed its formal test and is reported as exploratory.

Instrumentation. All significance statements use one instrument: the document-cluster paired bootstrap (10,000 draws, seed 0, percentile intervals; equal-cell pooling), unit-tested alongside the gate and pairing logic. Evaluation pins are sha-256 checksummed and byte-verified before use; the screen pin is a strict prefix of the confirmatory pin, and independent-extension analyses use the 50 new documents per cell only. Figures are generated from the verdict artifacts by a single script; nothing is hand-entered. Per-query, per-arm rows, frozen-bar metadata, verdict files, and the adversarial review trail are retained and will be released with the code. The descriptive re-slices in Appendix C.3, the realized-budget figures, the amortized-latency column of Table S10, and the budget sweep of Appendix B.6 are additional reuses of the development pin and carry no pre-frozen bars.

Sealed-test protocol. The 80-document confirmatory pin has been reused across this study and is therefore development-only. The fresh sealed set (zero overlap with the pin; 45 MMLB documents / 244 queries, plus 73 LDU documents left unevaluated as out of scope) was evaluated *exactly once* under a guarded, frozen harness: the confirmatory bar (complete-case document-clustered CI lower bound > 0, with an adversarial worst-case-imputation fallback binding only under any missingness), the corrected-loader comparator, the 5% budget, the seed, and sha-256 hashes of all eight code components plus the sealed pin were

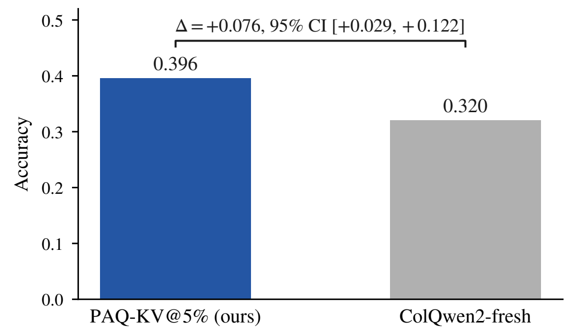

Figure S1: The sealed out-of-sample confirmation on MMLB: accuracy of PAQ-KV@5% and ColQwen2-fresh on the fresh, disjoint 45-document, 244-query set, evaluated exactly once under the frozen harness (§4.4 of the main paper). The bracket reports the paired difference with its document-clustered bootstrap 95% CI; only this comparison was run on the sealed set.

frozen before the run; the harness refuses the sealed pin absent an explicit authorization flag and a typed confirmation, aborts on any component-hash mismatch, and runs a blind non-scoring preflight that can abort without consuming the test. The single run produced the strongest claim in this paper: PAQ-KV@5% 0.396 vs. ColQwen2-fresh 0.320, $+0.076$ $[+0.029, +0.122]$ (244/244 complete, zero skips, robust to adversarial imputation; §5.2; Figure S1). The verdict, per-query rows, and a provenance manifest (pin and component hashes, resolved model revisions) are retained and released with the code.

Extended-benchmark overlap audit. The two added corpora are corpus-adjacent to ours: MMDocIR was sourced from existing DocVQA collections including MMLB, and MMDocRAG was built over the MMDocIR document pool. A document-name audit against our pins found substantial overlap, including, name for name, the entire sealed MMLB cell inside MMDocIR's pool. Every document overlapping either the development or the sealed pins was excluded from the new pins before any experiment ran, so the sealed reserve remains untouched at the document level; only document names were read from the sealed pin file, and the audit artifact is retained. The MMDocRAG pin's protocol deviations are disclosed here: 62 qualifying documents is below the study's 80-document floor; queries are capped at 12 per document by seed-0 sampling (585 of 1,160 kept), a cap the main pins never needed; pages render at 144 DPI, matching the existing pins' pixel scale; gold pages are the union of page identifiers over each query's gold quotes, with the 0-based convention verified by text-fragment matching on 32 of 32 probes; and answers are scored by the study's shared document-VQA evaluator. Because 28 of the 62 documents hit the twelve-query cap, we also decoded the 575 uncapped remainder queries on the same documents: the PAQ-KV-versus-ColQwen2 contrast is $+0.035$ $[+0.019, +0.060]$ on the remainder and $+0.046$ $[+0.032, +0.064]$ pooled, so the capped cell did not manufacture the result. The MMDocIR-QA cell reuses the MMDocIR selection-quality pin and scores against the benchmark's reference answers with the same shared evaluator. The extended cells reuse the study's frozen primitives (prefill, preview, selection, decode, the corrected ColQwen2 loader, the shared scorer, and the document-cluster paired bootstrap) but froze no bars, ran each cell once, and constructed the pins on the day of the runs. Two follow-ups closed this audit trail. A sealed-style MMDocRAG reserve was investigated and is structurally empty: none of the 41 remaining fresh documents qualify under the pin protocol (30 exceed the page ceiling, 11 lack three gold-bearing queries), so the MMDocIR-QA cell of §5.3 serves as the independent corroboration lane instead. And a training-distribution audit of ColQwen2, from its published training mixture, found no confirmed document-level overlap with any of the four evaluation corpora; MMDocIR's training split provably shares source datasets with ColQwen2's mixture, but our cells draw only on its evaluation pool, and web-crawled slices are unverifiable in both directions, so we claim neither contamination nor proven disjointness.

A.3 Methods-Integrity Detail

This section expands the summary of §4.4.

Per-query cache pollution. An independent code review found a real bug in the per-query scoring loop used across this study: the runtime mutates the retained cache in place, and the loop reused the document prefill across a document's queries without restoring it, so query q_i scored while attending to $q_0 \dots q_{i-1}$. The fix, cropping the cache after every per-query preview, is now applied everywhere, and its impact was quantified rather than assumed. On a 24-document, 150-query clean-versus-polluted re-run, the key isolation contrast is $+0.264$ clean versus $+0.284$ polluted, since the contrast largely cancels the bug when both arms share the same scorer. Every gate and headline number in this paper post-dates the fix, including the confirmatory isolation gate of §5.1, which was re-executed in full on post-fix code with PAQ-KV as the fresh arm and verified at runtime (cache-length assertions after every preview and decode, plus an order-invariance probe per cell that recomputes a query's preview after all other queries and requires bit-identity). The one surviving pre-fix artifact is the historical page-level panel of Appendix B.7, whose absolute fresh-arm accuracy carries ≈ 0.04 inflation; it is labelled as historical and no claim rests on its absolute levels. Appendix C.2 quantifies the pollution mechanism itself.

Baseline-loader integrity. A second independent review caught a defect in the *comparator*, not in our method: the cached ColQwen2 baseline had run with its trained final retrieval-projection LoRA ($\mathrm{custom\_text\_proj}$) silently zeroed. $\mathrm{vidore/colqwen2\text{-}v1.0}$ is a PEFT adapter, and under our peft/transformers versions a key remap inserted $\mathrm{language\_model}$ into the LoRA target path. The remap is correct for the language-model layers but wrong for the top-level $\mathrm{custom\_text\_proj}$, so the checkpoint's projection-LoRA tensors landed at a non-matching path and the final projection ran with base weights only. Because the other adapter layers still loaded, the baseline stayed functional and well above random, which masked the defect until we inspected the load report and the projection weights directly. We patched the loader, added a post-load assertion that the two $\mathrm{custom\_text\_proj}$ tensors match the checkpoint, and re-scored ColQwen2. The correction is immaterial to the main result. Fixed-versus-broken gold-page recall at the 5% budget is 0.539 versus 0.534. The top-5% page selections overlap at Jaccard 0.947, with only 34 of 444 queries changing any selected page. Re-reading those changed queries moves fresh-read accuracy by -0.0002 (0.3282 corrected versus 0.3284 broken), and the improvement over ColQwen2 is $+0.077$ against both loaders (LB_{95} $+0.040$ corrected, $+0.039$ broken), so the main result is not loader-inflated. The corrected loader is used for the main MMLB comparison (development and sealed), the MMDocRAG cell, the systems measurement, and, as of the same-protocol LDU close-out, the LDU comparison as well (top-5% selections overlap at Jaccard 0.968; 25 of 463 queries changed selection and were re-read; the comparator moves from 0.5124 to 0.5121 and the contrast is $+0.019$ against both loaders, so the correction is immaterial there too). Only the historical page-level cohort ever ran with the earlier loader; its panel table (Appendix B.7) omits those retrieval rows, and the page-level narrative of Appendix C.5 quotes them with disclosure.

B Complete Experimental Results

B.1 Key Paired Contrasts

Table S2 reports the paired contrasts summarized in Figure 3b.

B.2 Notes on Table 1

SnapKV-fresh scores pages from the same full-resolution prefill PAQ-KV uses, which roughly doubles it over the historical coarse-preview panel of Appendix B.7; we treat the same-cohort form as the fair one. BM25 text channels differ by corpus: LDU has no born-digital text layer, so its cell uses a tesseract OCR channel (98.1% of pages recovered), the MMDocIR-QA cell uses the benchmark's own OCR text field (99.5% coverage), and the other cells use born-digital text. The LDU ColQwen2 cell uses the corrected loader (§4.4). MMLB decodes 444 queries; paired contrasts use the 440 with cached references (the 4 $\mathrm{mi\_phone}$ rows lack cached references). Realized budgets on MMLB: PAQ-KV attends a mean 6.0% of the prefill context (median 6.0, range 5.5 to 7.5%), including the always-attended non-visual prefix; ColQwen2 keeps whole pages, a mean 4.6% of pages under whole-page rounding.

The roughly one-point realized overshoot cannot explain the main comparison. First, the units are not symmetric: part of PAQ-KV's 6.0% is the non-visual scaffold rather than document evidence, and the comparator's reader receives the same scaffold and question uncounted. Second, evidence delivery runs the other way: at the same nominal budget the retriever hands the reader substantially more gold-evidence tokens (coverage-weighted recall 0.539 versus 0.344; Appendix C.3) and still loses the accuracy comparison. Third, the budget sweep orders against a realized-budget account:

Benchmark	Δ [95% CI], PAQ-KV minus ColQwen2-fresh	full-context	gold-page oracle	Bar
LDU	$+0.019\,[-0.019,+0.061]$	$-0.014\,[-0.031,+0.003]$	$-0.078\,[-0.116,-0.040]$	NOT MET
MMLB	$+0.077\,[+0.040,+0.115]$	$+0.009\,[-0.022,+0.040]$	$-0.026\,[-0.059,+0.008]$	**MET**
MMLB (sealed)	$+0.076\,[+0.029,+0.122]$	–	–	**MET**
MMDocRAG	$+0.057\,[+0.040,+0.076]$	$+0.007\,[-0.004,+0.018]$	$+0.006\,[-0.009,+0.022]$	none frozen
MMDocIR-QA	$-0.043\,[-0.065,-0.011]$	$-0.013\,[-0.049,+0.012]$	$-0.095\,[-0.158,-0.022]$	none frozen

Table S2: Key paired contrasts for Table 1: mean paired difference Δ with its document-clustered bootstrap 95% confidence interval (10,000 draws; §4.2). The pre-registered per-cell bar (final column) passes when the CI lower bound of the ColQwen2-fresh contrast exceeds zero; dashes denote arms not run on the sealed set. On LDU PAQ-KV is below full context, with an interval that includes zero, and below the gold-page oracle, with an interval that excludes zero; on MMLB and MMDocRAG it is statistically indistinguishable from both references. On MMDocIR-QA the trained retriever significantly outperforms PAQ-KV, which remains indistinguishable from its own full-context read; §5.3 interprets the four-cell pattern. The MMDocRAG and MMDocIR-QA cells are development-grade, so no bars were frozen there.

the margin is largest at the 2.5% nominal budget, where the realized fractions essentially coincide (3.5% of context versus 3.6% of pages), and smallest at 10%, where PAQ-KV's realized excess is largest (10.9 versus 8.6%; Table S6).

B.3 Confirmatory Isolation Gate Table

Table S3 gives the full confirmatory isolation-gate values behind §5.1 and Figure 3a.

B.4 Cross-Reader Results in Full

Table S4 gives the full cross-reader values behind Figure 3c. Every port had to pass the same all-true-mask decode-identity gate (teacher-forced argmax parity against the native causal decode, plus a restrictive mask that changes the output, on documents spanning at least 30 pages) before any accuracy number was read. Against their own full context on LDU, Qwen3-VL-4B is indistinguishable ($-0.006\,[-0.027,+0.017]$) while Qwen2.5-VL-7B and InternVL3.5-8B fall short ($-0.060\,[-0.100,-0.021]$ and $-0.077\,[-0.109,-0.047]$); InternVL3.5-8B is also below its own full context on MMLB ($-0.076\,[-0.113,-0.041]$).

InternVL tuning study. A pre-registered development-only tuning study on InternVL3.5-8B (budgets $\{5,10,20,30\}$%, four attention read-outs, tuned on 40 documents and evaluated on the held-out 40) found budget to be the only effective lever: the deficit narrows monotonically to -0.001 at a 30% budget, but the lower bound never clears zero at any budget, and the tuned read-out adds nothing beyond budget ($+0.016$, LB_{95} -0.014, paired against the default), so the negative is not an artifact of the 5% budget or of the default attention read-out.

Incompatible candidates. Three further candidates, Molmo-7B-D (Deitke et al. 2025), MiniCPM-V-4.5 (Yu et al. 2026), and PaliGemma-2 (Steiner et al. 2024), could not run the method under the pinned software stack (remote-code version drift for the first two; an $\approx$8K context that cannot hold a multi-page document for the third) and are reported as incompatible rather than forced. This maps the method's practical deployment envelope: a version-current implementation exposing hookable attention and an honored 4D additive mask, and enough context to prefill a whole document.

B.5 Cross-Reader Baseline Panel

The cross-reader results of §5.3 compare PAQ-KV against ColQwen2-fresh and full context on the three second readers. This appendix adds the remaining arm family at the same $b=5\%$ budget (Table S5), with page-set semantics identical to the Qwen3-VL-8B panel: the same cached CLIP and ColQwen2 page scores, the same commit-once aggregation over the first $M=\min(3,k)$ queries, the same seed-0 random and truncation controls, and the gold-page oracle read directly. The read path is identical to each port's ColQwen2 comparator arm, so every number is within-reader comparable. SnapKV arms are computed from each reader's *own* question-preview attention, aggregated to pages with the panel's pooling, so SnapKV-fresh and SnapKV-stale are genuinely per-reader rather than proxied from the development reader. All six reader-cell runs are complete (907 queries by seven arms per reader, zero skipped rows), and everything in this appendix is development-grade. The same arm family was also executed on the development reader itself; those same-cohort values replace the historical panel's baseline rows in Table 1.

Three regularities stand out. First, PAQ-KV stays ahead of every training-free baseline on every reader, including the readers where it loses to the trained retriever: SnapKV-fresh, the strongest such baseline, sits significantly below PAQ-KV in all six reader-cell combinations. Second, SnapKV-fresh here is exactly the reader's own question-preview attention selecting pages, so its absolute level reads out each reader's attention-as-retriever quality; it falls across readers in the same order as the recovery gradient of §5.3, so the selection signal itself degrades in that order, independently of the block machinery. Third, committing either scorer once collapses it on every reader, so the multi-query staleness penalty is a property of the regime rather than of the development reader. On InternVL3.5-8B the gold-page oracle additionally exceeds the reader's own full-context accuracy, so that reader's bottleneck is finding evidence rather than converting it: where the host's attention retrieves poorly, a stronger

Contrast at b=5%	LongDocURL	MMLongBench-Doc	pooled (equal-cell, secondary)
fresh − same-scorer commit-once	$+0.211$ $[+0.136, +0.289]$	$+0.137$ $[+0.103, +0.169]$	$+0.174$ $[+0.132, +0.215]$
fresh − gold-informed static	$+0.230$ $[+0.178, +0.276]$	$+0.213$ $[+0.160, +0.266]$	$+0.221$ $[+0.184, +0.256]$
arm means (fresh / commit-once / static)	0.531 / 0.320 / 0.301	0.405 / 0.268 / 0.192	—
Verdict	REVERSIBILITY ISOLATED (all CI lower bounds > 0, both cells)		

Table S3: Confirmatory below-page isolation at b=5% on all 907 pinned queries (accuracy; PAQ-KV as the fresh arm; postfix code; document-cluster paired bootstrap; contrast rows report Δ with 95% CIs). The two-comparator bar was frozen and committed before the run; the per-cell contrasts are the pre-registered primary, the pooled column is secondary.

selector recovers real accuracy that PAQ-KV leaves on the table.

Quantitatively: SnapKV-fresh sits below PAQ-KV by -0.061 $[-0.096, -0.025]$ (MMLB) and -0.083 $[-0.144, -0.031]$ (LDU) on Qwen3-VL-4B, by -0.143 on both Qwen2.5-VL-7B cells, and by -0.099 and -0.142 on InternVL3.5-8B; its absolute level falls from 0.303 to 0.127 to 0.089 across the three readers on MMLB, and from 0.413 to 0.252 to 0.077 on LDU. On InternVL3.5-8B the gold-page oracle exceeds full context by $+0.052$ $[+0.013, +0.094]$ on MMLB and $+0.045$ $[+0.007, +0.088]$ on LDU.

B.6 Full-Pin Budget Sweep

The improvement is not a 5%-only artifact, and on the full pin the earlier screen-only caveats resolve favorably. PAQ-KV's full-pin budget curve is monotone (0.389/0.404/0.405 at 2.5/5/10%), so the mild non-monotonicity previously observed on the 157-query screen subset (0.397/0.404/0.398) was small-sample noise. Against a ColQwen2 comparator re-read at the *matched* page budget (Table S6), PAQ-KV is clearly ahead at 2.5%, ahead by the pre-registered margin at 5%, and ahead on point estimate at 10% with an interval that just includes zero: the margin narrows as budget grows, because the retriever gains more from additional whole pages than PAQ-KV gains from additional blocks. Against the fixed ColQwen2@5% comparator, every PAQ-KV budget remains ahead ($+0.064$, $+0.077$, $+0.078$ at 2.5/5/10%, all lower bounds positive; 440 paired queries).

B.7 Historical Page-Level Panel

Table S7 reports the historical page-level panel at the gate budget on the confirmatory pin. Its fresh arm predates the cache fix of §4.4 ($\approx$0.04 absolute inflation); it is retained because it is the only executed home of the query-blind eviction family, which sits at or below random gold-page recall on these cells (§4), and as context for Appendix C.5. The panel's retrieval and reference arms (ColQwen2, CLIP, full-context, oracle) are omitted from the table: they ran with the uncorrected ColQwen2 loader and are superseded by the same-cohort corrected rows of Table 1; where the historical values are needed for the page-level narrative they appear, disclosed, in the prose of Appendix C.5. No headline claim rests on this panel's absolute levels. Secondary pooled statistics: the fresh page-level selector exceeds the best arm of the eviction family (SnapKV-stale) with min-over-family LB_{95} $+0.202$; fresh $-$ SnapKV-fresh $=$ $+0.094$ $[+0.069, +0.122]$; fresh $-$ oracle $=$ -0.199 $[-0.239, -0.161]$. Gold-page recall at b=5% (LDU / MMLB): fresh page selector $0.494/0.443$; gold-informed static $0.381/0.373$; same-scorer commit-once $0.137/0.148$; ColQwen2-fresh $0.664/0.536$.

B.8 MMDocIR: Below-Page Selection Quality

MMDocIR (Dong et al. 2025) provides expert gold at *layout* granularity: paragraphs, tables, figures, and charts as boxes on the page. This is ground truth at exactly PAQ-KV's below-page granularity, which neither QA benchmark offers. On a fresh-documents-only pin (80 documents, 538 queries, 641 gold layouts; the same exclusion audit as Appendix A.2), we score PAQ-KV's selected 64-token blocks geometrically against the gold boxes: block token spans are mapped through the reader's own patch grid to page-pixel rectangles, and we measure gold-layout area coverage, layout hit rate, and gold-page recall at $b \in \{5, 10\}$% against random-block, truncation-block, and page-granularity comparators at matched token budget (Table S8). No decoding is involved, so the analysis isolates the selector from QA conversion entirely. The same pin also yields the MMDocIR-QA accuracy cell of §5.3; there, the benchmark's reference answers skew long, which depresses absolute levels under the study's short-answer scorer without affecting the paired contrasts (a 30-probe scorer-fit check preceded the run, and one query with an empty gold answer scores zero for all arms).

Three observations. The selector finds sub-page evidence: at a realized 5.3% of context, PAQ-KV's blocks cover 0.689 of gold-layout area, where random blocks at the same budget cover 0.056. This is the study's first conversion-free measurement of the selection signal against human-annotated sub-page ground truth. Below-page granularity buys reach rather than raw area: the page-granularity arm, using identical attention scores aggregated to pages, covers the same gold area but needs a realized 7.7% of context to do it under whole-page rounding, and at that higher spend still touches fewer distinct gold layouts ($+0.140$ $[+0.097, +0.170]$ in PAQ-KV's favor) and fewer gold pages. Blocks spread the same budget across more of the question's evidence, which is the behavior the cross-page QA advantage of Appendix C.3 suggested, now measured geometrically. Finally, selecting every gold page in full would cost a realized 5.6% of context on this pin, essentially PAQ-KV's own spend, so a perfect page-level selector could reach full coverage at matched budget; PAQ-KV recovers 69% of that area with no supervision, from the reader's own attention, and the remaining gap is the

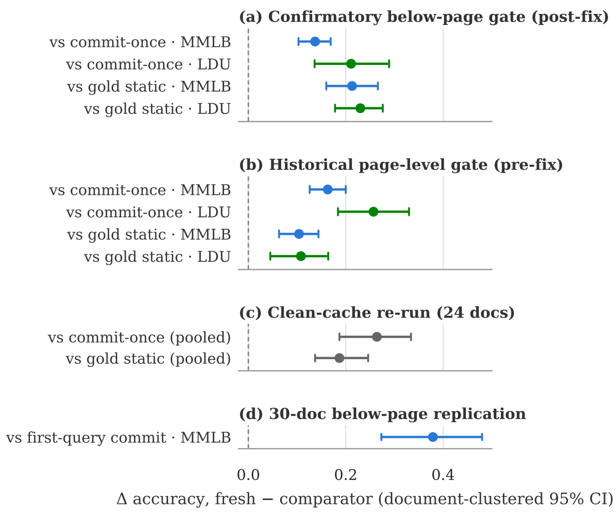

Figure S2: Document-clustered two-sided 95% intervals for every executed isolation contrast (blue MMLB, green LDU, gray pooled over both cells). **(a)** The confirmatory below-page gate (PAQ-KV as the fresh arm, full 80-document pins, post-fix code; identical to Figure 3a). **(b)** The historical page-level gate (pre-fix cohort). **(c)** The clean-cache re-run of Appendix A.3. **(d)** The 30-document below-page replication, whose committed arm is the weaker first-query commit (Appendix C.1). Positive differences favor per-query reselection; every interval excludes zero.

selection headroom that the oracle contrasts of Appendix B.5 price in accuracy terms.

C Additional Analyses

C.1 Mechanism Checks in Detail

On the 30-document screen (distinct from the 80-document confirmatory cohort), the MMLB result survives every mechanistic adversarial check: it is query-dependent (PAQ-KV − random-block selection $= +0.329$ $[+0.238, +0.418]$), query-specific (PAQ-KV − wrong-question selection $= +0.698$ $[+0.603, +0.784]$, where the collapse supports query-specificity), reversible (PAQ-KV − commit-once $= +0.379$ $[+0.273, +0.480]$; this screen control commits the *first query's* selection, a weaker committed arm than the confirmatory gate's first-three-query aggregate, which is why its margin exceeds the confirmatory $+0.137$), and below-page granularity helps over page level ($+0.054$ $[+0.011, +0.102]$). These are development-pin screen analyses with disclosed caveats: the screen budget-curve monotonicity gate failed on this cell (0.398 at 10% vs. 0.402 at 5% on the 157-query screen subset), though the subsequent full-pin sweep is monotone and attributes that failure to small-sample noise (Appendix B.6), and the effect they dissect is separately confirmed out-of-sample on the sealed set (§5.2).

C.2 Additional Reversibility Detail

Decode-identity audit detail. The six audited document-query pairs of §3 span 32K to 122K tokens, a stress range deliberately wider than the evaluation pin. An earlier single greedy-token divergence traced to the generation loop's mask-independent fast-path kernel, not to the mask.

Robustness checks behind §5.1. First, an independent extension: because the screen pin is embedded in the confirmatory pin, we re-ran the statistics on the 50 new documents per cell only ($n = 586$ queries), where fresh over commit-once is $+0.180$ $[+0.124, +0.237]$ and fresh over the static $+0.206$ $[+0.161, +0.249]$ (pooled, equal-cell). Second, granularity and history: the same two-comparator gate passed at page level on this pin (its pre-fix historical run: $+0.257$/$+0.163$ over commit-once), and the page-level contrast survives the clean-cache re-run of Appendix A.3 ($+0.264$, LB_{95} $+0.187$), so the effect is not specific to block granularity or to the fix. Third, scorer generality: the same fresh-over-stale collapse holds for a retrieval scorer, with ColQwen2 fresh at 0.423 versus stale at 0.178 (historical page-level cohort; query-micro; equal-cell pooling agrees). The pre-registered "commitment-pressure" gradient failed its formal interaction test with both intervals including zero.

Figure S2 summarizes the confirmatory below-page gate and its supporting replications. For the confirmatory gate, missingness is a non-issue: all 907 pinned queries completed with zero skips, so the pre-registered adversarial imputation guard (fresh scored 0, comparators scored 1 on any incomplete query) never binds. All three arms realize a matched ≈6.0% of context. Supporting analyses from the historical page-level run (pre-fix cohort, labelled): (i) *answerable subset* (secondary, since it conditions on a model outcome): on queries where oracle-evidence scores exactly 1 ($n = 407$), fresh 0.583 vs. commit-once 0.182 ($+0.395$ $[+0.324, +0.460]$) and vs. the static 0.382 ($+0.204$ $[+0.135, +0.266]$); (ii) *budget curves*: the fresh page-level selector dominates both static comparators at every budget in $\{2.5, 5, 10\}$% on both cells.

Cache-pollution quantification. The in-place cache bug of §4.4 was quantified before being fixed: on a 6-documents-per-cell diagnostic (87 query-rows), the bug inflates gold-page recall by $\approx +0.05$ for a fixed scoring head and shifts selections (mean top-B Jaccard clean vs. polluted 0.82, drifting from 1.00 at q_0 to 0.61 at q_6), as expected for cumulative contamination. The recall gate behind the PAQ-T numbers was regenerated clean before any accuracy read (clean held-out expert recall 0.683/0.626 vs. polluted 0.674/0.628), and the clean cross-validation picks a more robust head. Absolute fresh-arm accuracy in the pre-fix confirmatory run carries ≈0.04 inflation (clean − polluted -0.041 $[-0.077, -0.003]$ on the 24-document check); the isolation contrasts cancel the bug and stand.

C.3 Selection Anatomy on MMLB

These are descriptive re-slices of the retained per-query development-pin rows (444 decoded / 440 paired, corrected ColQwen2 loader throughout); no pre-frozen bar attaches to them.

Gold-page recall vs. accuracy. At the nominal 5% budget, two recall notions bracket what each method hands the reader. Coverage-weighted (the fraction of gold-page *tokens* the reader may attend; the budget-comparable notion, since ColQwen2 keeps whole pages): ColQwen2 0.539 vs. PAQ-KV 0.344, a deficit of -0.195 $[-0.236, -0.152]$ for PAQ-KV. Any-block-touch (a gold page counts as recalled if any selected 64-token block overlaps it): PAQ-KV 0.912, but this notion flatters block granularity, since PAQ-KV's selections scatter over a median of 10 pages per query. So the retriever delivers substantially more gold-evidence tokens at matched budget, touches fewer of the right pages, and still loses the accuracy comparison by $+0.077$ (Table 1): recall-optimized selection is not accuracy-optimized selection, consistent with the sharp-scorer negative result (Appendix C.4).

Evidence-type stratification. Table S9 stratifies the main paired comparison by the benchmark's evidence annotations. The cross-page stratum roughly doubles the single-page contrast ($+0.109$ vs. $+0.054$), and the source strata whose intervals exclude zero (chart, table, layout) are those whose evidence a page-level retriever must fetch whole even when only a small region answers the question.

C.4 Negative Results

Per-head mixture transfer gate. No LoRA adapter was ever trained: this pre-frozen check failed first and stopped that line of work. We fit a logistic per-head mixture over all $36 \times 32 = 1{,}152$ (layer, head) question-to-page attention channels on off-benchmark provenance-checked data. The transfer gate fails: mixture gold-page recall is 0.652/0.594 vs. 0.676/0.616 for the single head, with pooled mixture $-$ single $= -0.023$ $[-0.037, -0.008]$ (907 queries), so the mixture overfits and this direction was closed. The best zero-benchmark-contact head (L21.H18) still coincides with the benchmark-supervised expert layer.

Sharp expert head as block scorer. Replacing PAQ-KV's diffuse mean-attention block scorer (V0) with the sharp head L21.H18 (V1) fails the pre-frozen gate: on LDU it only nudges the point estimate (from 0.5315 to 0.536; V1 $-$ ColQwen2 $= +0.024$ $[-0.012, +0.066]$) and does not separate from V0 (the V1$-$V0 LB_{95} is -0.011, n.s.), while on MMLB it hurts (from 0.402 to 0.381 on the screen documents). As elsewhere in this study, recall-optimized selection is not accuracy-optimized selection.

C.5 The Page-Level System in Detail

The name PAQ abbreviates "Prefill once, Ask many Queries"; unsuffixed PAQ denotes the earlier page-level router (v1) detailed in this appendix, PAQ-T its expert-head variant, and PAQ-KV the below-page KV-mask method of the main text. All numbers in this section are on the 903-query confirmatory cohort (page-level arms), where ColQwen2-fresh scores 0.512/0.328 (LDU/MMLB) with the uncorrected loader; it is distinct from the below-page cohort used in the main comparison (where the corrected-loader ColQwen2-fresh MMLB is 0.3282, §4.4). The loader correction is immaterial (§4.4), so the page-level parity conclusions are unaffected.

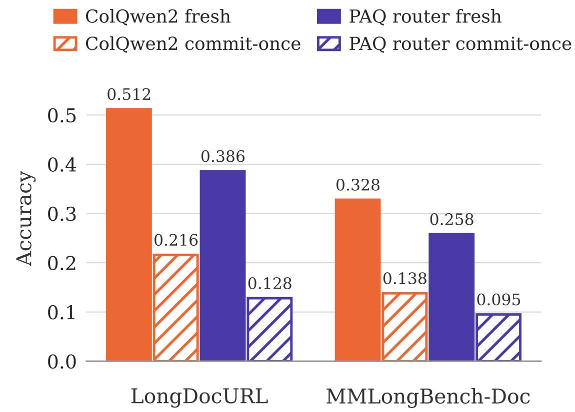

Figure S3: Two observations from the page-level system, per cell (solid bars = per-query fresh selection, hatched open bars = the same scorer committed once): (i) ColQwen2-fresh outperforms the page-level attention router (a loss of roughly 10 points for the router); (ii) *both* scorers collapse when committed once, so reversibility is a general property of the multi-query regime rather than being scorer-specific. PAQ here is the earlier page-level router of this appendix, not PAQ-KV; the below-page PAQ-KV comparison at matched budget is in Table 1. Historical cohort, uncorrected ColQwen2 loader (disclosed above).

The page router (PAQ v1). Pages are rendered at coarse resolution ($\approx$3.5K visual tokens total, i.e. $\approx$70 tokens/page on a 50-page document) and prefilled once; per query, the question rows' attention (mean over heads, layers, and question tokens) scores pages; the top pages at budget b are re-rendered at full resolution and read. This is page routing rather than KV compression, and it claims no memory savings.

Comparison against trained retrieval. At $b = 5\%$, ColQwen2-fresh scores 0.512/0.328 vs. PAQ 0.386/0.258: pooled -0.099 $[-0.129, -0.072]$, a loss of roughly 10 points that holds on every stratum we examined (Figure S3). Gold-page recall is consistent with a coverage contribution: ColQwen2-fresh 0.664/0.536 vs. PAQ 0.494/0.443. Both scorers collapse when committed once (ColQwen2 stale 0.216/0.138), reproducing the reversibility pattern with a second scorer family.

PAQ-T: expert-head recall and the parity accuracy evaluation. Scoring with a single expert (layer, head) chosen by held-out-document cross-validation (CAPS's technique) lifts held-out gold-page recall@5% to 0.683 (n=451) / 0.626 (n=425), meeting or exceeding ColQwen2's in-harness 0.664/0.536; mean aggregation at higher scoring resolution moves almost nothing ($0.494/0.443 \rightarrow 0.500/0.452$), so the head, not the resolution, is the lever. All selected fold heads sit in layer 21. The pre-registered accuracy evaluation (bars frozen before the run; same reader, render policy, and scorer as the cached baseline rows; only the selected page set differs) lands at parity: PAQ-T 0.518/0.357 vs. ColQwen2-fresh

0.512/0.328; deltas $+0.006$ $[-0.018, +0.030]$ and $+0.028$ $[+0.001, +0.055]$; pooled $+0.017$ $[-0.000, +0.035]$. This is non-inferior under the frozen lower-bound margin (> -0.03) on both cells but not the pre-registered improvement criterion (all lower bounds > 0), and we do not present the marginal per-cell interval as evidence of superiority. The initially better-looking complete-case verdict (pooled $+0.023$, LB_{95} $+0.005$) was blocked in review for informative missingness: the $\approx$4% of queries whose high-resolution scoring ran out of memory had cached baseline accuracy 0.62, far above the cell means; a conservative reduced-resolution recovery of 31/35 missing rows (degrading only the PAQ-T arm, never the baseline) erased the pooled edge. Two caveats carry over: head selection is cross-validated *benchmark-supervised* model selection (not zero-shot; the externally selected L21.H18 of the negative transfer-gate ablation later removed this caveat from the head's existence, though not from its benchmark-tuned use), and the expert-head technique is CAPS's, so the residual novelty at page level is only the amortized one-prefill-many-queries architecture.

C.6 Systems Measurements in Detail

We measured the deployment profile of the true PAQ-KV path (below-page masked decode over the *retained* full-document KV cache) against ColQwen2-fresh and full context on an 8-document, 31-query systems subset of the pin (single GPU, frozen configuration; Table S10). PAQ-KV is roughly $2.25\times$ slower per online query than the retriever pipeline and retains a multi-gigabyte per-document cache where the retriever stores a megabyte-scale page index, but it is roughly $11\times$ faster than re-reading the document for every question, and the one-time prefill amortizes after about one query per document. PAQ-KV therefore does not reduce resident KV memory; it trades retained cache state for reversible query-specific selection. The efficiency claim is simplicity and deployment shape, not speed: no separately trained retriever checkpoint, no stored vector index, one deployed model, and no repeated full-document prefills.

D Additional Limitations Detail

LDU conversion-bound decomposition. A re-analysis of the executed rows suggests the shortfall there lies largely past selection: of the 463 rows, $\approx$39% fail under both PAQ-KV and the gold-page oracle (zero even when handed the native-resolution gold pages), only $\approx$10% are oracle-right/PAQ-KV-wrong (the selection-capturable pool), and 2.2% are PAQ-KV-right/oracle-wrong, so the oracle is not a strict ceiling; that no selection method can rescue these queries is an observation about this dataset, not a general rule. Native resolution already saturates the prefill on 79/80 documents, and large documents are a flat, low-leverage half. Among pure-selection levers, only a marginal oracle-chooser union passes (LB_{95} $+0.003$; not deployable); only a PAQ-KV$\cup$ColQwen2 hybrid passes comfortably (LB_{95} $+0.061$), and we exclude it as not a standalone method. We therefore offer the dispersed-vs-conversion-bound split as a hypothesis from two datasets.

Base-model ceiling counts. A large fraction of queries receive no credit even given exactly the annotated gold pages: the oracle-evidence arm scores 0 on $317/903 = 35.1\%$ of confirmatory queries and a full 1 on only 45.1%. Pairing controls for shared query difficulty; the oracle is not a strict ceiling (a selected page set can score above the annotated gold set on individual queries).

Remaining protocol caveats. The commit-once aggregate uses the first $M = \min(3, k)$ queries in frozen arrival order (order-permutation and M-sensitivity checks remain to be run; the order-free static partially covers this), and the page-level confirmatory run predates the cache fix (§4.4); its isolation *contrasts* were verified robust, but its absolute fresh-arm levels carry ≈ 0.04 inflation on the 24-document check.

Method	LDU	MMLB
Qwen3-VL-8B (development reader; $\sim$69K ctx)		
PAQ-KV@5% (ours)	**0.5315**	**0.4042**
ColQwen2 (matched)	0.5121	0.3282
full-context (ref.)	0.5456	0.3965
Δ [95% CI]	$+0.019\,[-0.019, +0.061]$	$+0.077\,[+0.040, +0.115]$
Qwen3-VL-4B (same architecture; $\sim$69K ctx)		
PAQ-KV@5% (ours)	**0.4966**	**0.3640**
ColQwen2 (matched)	0.4908	0.3336
full-context (ref.)	0.5024	0.3480
Δ [95% CI]	$+0.006\,[-0.028, +0.043]$	$+0.030\,[-0.004, +0.065]$
Qwen2.5-VL-7B (same family, prior gen.; $\sim$52K ctx)		
PAQ-KV@5% (ours)	0.3953	**0.2698**
ColQwen2 (matched)	**0.4728**	0.2623
full-context (ref.)	0.4558	0.3237
Δ [95% CI]	$-0.078\,[-0.116, -0.037]$	$+0.007\,[-0.037, +0.049]$
InternVL3.5-8B (cross-family, Qwen3 LM; $\sim$36K ctx)		
PAQ-KV@5% (ours)	0.2186	0.1876
ColQwen2 (matched)	**0.3081**	**0.2542**
full-context (ref.)	0.2961	0.2637
Δ [95% CI]	$-0.090\,[-0.127, -0.057]$	$-0.067\,[-0.106, -0.029]$

Table S4: Cross-reader generality at the 80-document development pins (LDU = LongDocURL, 463 queries; MMLB = MMLongBench-Doc, 444 queries). Within each reader block, PAQ-KV (the reader's own attention selects $\approx$5% of blocks) is compared with the same reader reading the top-5% pages ranked by the cached ColQwen2 scores, so every contrast is within-reader at matched budget; the reader's own full context is a reference, not a competitor. Bold marks the better of the two matched-budget arms per column by point estimate; significance is carried by the contrast rows, which report the PAQ-KV-minus-ColQwen2 difference with its document-clustered 95% CI. Block headers give the mean realized MMLB prefill length: the two non-Qwen3-VL readers run below the $\approx$69K budget because of their context ceilings, so only within-reader comparisons are interpreted. The Qwen3-VL-8B block repeats Table 1; its MMLB improvement is additionally confirmed on the sealed set (§5.2). Qwen3-VL-4B reuses byte-identical input geometry (verified on all 80 documents), so that block is a pure size ablation. This table carries the paired contrasts and their CIs; Table S5 extends the same second readers with the full training-free arm family (point estimates only).

Arm	LDU	MMLB
Qwen3-VL-4B (same architecture)		
PAQ-KV@5% (ours; cached)	0.4966	0.3640
ColQwen2-fresh (cached)	0.4908	0.3336
full-context (ref.; cached)	0.5024	0.3480
SnapKV-fresh	0.4133	0.3034
SnapKV-stale	0.1332	0.0873
CLIP-fresh	0.1545	0.1805
ColQwen2-stale	0.2105	0.1576
random	0.0684	0.0296
truncation	0.0322	0.0423
gold-page oracle (ref.)	0.5839	0.4373
Qwen2.5-VL-7B (same family, prior gen.)		
PAQ-KV@5% (ours; cached)	0.3953	0.2698
ColQwen2-fresh (cached)	0.4728	0.2623
full-context (ref.; cached)	0.4558	0.3237
SnapKV-fresh	0.2519	0.1265
SnapKV-stale	0.0574	0.0342
CLIP-fresh	0.1591	0.1354
ColQwen2-stale	0.2047	0.1245
random	0.0609	0.0253
truncation	0.0343	0.0394
gold-page oracle (ref.)	0.5380	0.3219
InternVL3.5-8B (cross-family, Qwen3 LM)		
PAQ-KV@5% (ours; cached)	0.2186	0.1876
ColQwen2-fresh (cached)	0.3081	0.2542
full-context (ref.; cached)	0.2961	0.2637
SnapKV-fresh	0.0770	0.0889
SnapKV-stale	0.0537	0.0492
CLIP-fresh	0.1121	0.1498
ColQwen2-stale	0.1549	0.1068
random	0.0675	0.0377
truncation	0.0293	0.0385
gold-page oracle (ref.)	0.3408	0.3159

Table S5: Baseline arm family on the three second readers at the 80-document development pins (LDU = LongDocURL, MMLB = MMLongBench-Doc), $b = 5\%$. The PAQ-KV, ColQwen2-fresh, and full-context rows are the cached values from the cross-reader ports (identical to Table S4); the remaining arms are new runs with matched page-set semantics. Development-grade.

Budget	PAQ-KV	ColQwen2@b	Δ [95% CI]
2.5%	0.389	0.291	$+0.098\,[+0.058, +0.137]$
5%	0.404	0.328	$+0.077\,[+0.040, +0.115]$
10%	0.405	0.370	$+0.036\,[-0.001, +0.072]$

Table S6: Full-pin budget sweep on MMLB (80 documents; matched budgets, corrected ColQwen2 loader). The 5% row is the pre-registered main comparison (440 paired queries, Table 1); the 2.5/10% rows are descriptive (444 queries each; the pre-frozen bar was frozen at 5% only). Realized fractions: PAQ-KV 3.5/6.0/10.9% of context, ColQwen2 3.6/4.6/8.6% of pages.

Group	Arm	LDU	MMLB
ours (page level)	PAQ (fresh)	**0.386**	**0.258**
	attn_max (alt-agg)	0.337	0.152
isolation	same-scorer commit-once	0.128	0.095
	gold-informed static	0.278	0.154
reference	SnapKV-fresh	0.298	0.157
query-blind	SnapKV-stale	0.083	0.074
	H2O	0.045	0.039
	FastV (proxy)	0.045	0.031
	LOOK-M (→H2O)	0.045	0.039
	KVzip (proxy)	0.047	0.048
	random	0.068	0.022
	truncation	0.033	0.041

Table S7: The historical page-level panel: accuracy at b=5%, per cell (LDU = LongDocURL, MMLB = MMLongBench-Doc), on the 903-query confirmatory cohort (paq_verdict; pre-fix fresh arm, ≈ 0.04 absolute inflation). Query-aware arms marked fresh re-select per query; stale arms commit once per document. The cohort's uncorrected-loader retrieval rows and its full-context/oracle reference rows are omitted; the same-cohort corrected values are in Table 1 and the historical values, as needed, in Appendix C.5.

Arm	layout cov.	layout hit	page recall	realized b
nominal budget $b = 0.05$				
PAQ-KV blocks (ours)	**0.689**	0.834	0.903	5.3%
page level (same scorer)	0.693	0.694	0.694	7.7%
random blocks	0.056	0.141	0.449	5.3%
truncation blocks	0.079	0.087	0.095	5.4%
nominal budget $b = 0.1$				
PAQ-KV blocks (ours)	**0.778**	0.895	0.960	10.3%
page level (same scorer)	0.791	0.792	0.792	12.2%
random blocks	0.114	0.273	0.720	10.2%
truncation blocks	0.125	0.127	0.138	10.3%
whole gold pages (upper ref.)	1.000	1.000	1.000	5.6%

Table S8: Selection quality against MMDocIR's sub-page gold layouts on the 80-document fresh pin (538 queries). "Coverage" is the fraction of gold-layout area covered by selected blocks on gold pages; "hit" the fraction of gold layouts touched at all; "gold-page recall" the fraction of gold pages receiving any block. Development-grade.

Stratum	n	docs	PAQ-KV	ColQ.	Δ [95% CI]
Chart	102	35	0.409	0.311	$+0.098$ $[+0.025, +0.176]$
Figure	156	53	0.393	0.333	$+0.060$ $[-0.020, +0.143]$
Layout	47	32	0.490	0.338	$+0.152$ $[+0.032, +0.252]$
Pure text	144	52	0.356	0.308	$+0.048$ $[-0.010, +0.104]$
Table	120	38	0.353	0.258	$+0.096$ $[+0.033, +0.163]$
Single-page	256	74	0.526	0.471	$+0.054$ $[+0.001, +0.108]$
Cross-page	184	70	0.238	0.129	$+0.109$ $[+0.052, +0.170]$

Table S9: Paired PAQ-KV@5% − ColQwen2-fresh (corrected loader) by MMLB evidence stratum on the 440-query paired cohort (document-cluster paired bootstrap). Evidence-source labels are the benchmark's own and are multi-label (a query counts under each listed source); single- vs. cross-page splits on the number of annotated gold pages. The improvement concentrates on cross-page evidence and on chart/table/layout sources; these are descriptive development-pin strata, not pre-registered claims.

	extra params	per-doc state	cold st. /doc (s)	online /q (s)	amort. /q (s)	VRAM (GiB)
PAQ-KV (ours)	0	8.15 GiB	11.3	1.04	3.07	44.5
ColQwen2-fresh	2.246 B	11 MiB	2.5	0.46	0.91	21.8
full-context	0	n/a	n/a	11.59	11.59	44.6

Table S10: Deployment profile on an 8-document / 31-query systems subset (frozen configuration, single GPU). "Extra params" = additional retriever/selector parameters beyond the shared reader. "Cold start" is the one-time per-document cost: full-document prefill for PAQ-KV, page-embedding index build for ColQwen2. "Amort. lat./q" adds the cold start amortized at the development pin's observed 5.55 queries/document to the online latency (at the systems subset's own 3.9 queries/document the figures are 3.95/1.10/11.59 s). PAQ-KV's online latency decomposes into 0.19 s selection and 0.85 s masked decode; it is $\approx 2.25\times$ slower and far heavier in resident state than ColQwen2, and $\approx 11\times$ faster than full context (cold start amortized after $N \approx 1.07$ queries/document). PAQ-KV's online latency excludes ≈ 1.99 s/query of removable research-harness input reconstruction (component sum ≈ 3.03 s/query in the current implementation); mean generated lengths differ across arms (5.3/3.5/4.1 tokens for PAQ-KV/ColQwen2/full-context).

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